

SMC Premium/Discount Confluence Engine

v17.20

Completion Report — CRITICAL EQ Line Z-Order Fix + Navy Blue HUD

Date: 2026-05-16
Type: Visual-Only Update (ZERO logic/filter/alert architecture changes)
Plot Budget: 33/64 (unchanged)

CRITICAL FIX: EQ Line Z-Order (NOW VISIBLE)

Root Cause: The EQ (Equilibrium) line was drawn INSIDE the same code block as the P&D zone boxes. In Pine Script, filled boxes painted over the line because both were created in the same rendering pass. The EQ line was effectively invisible on the chart.

Solution Applied:

- 1. **Separated EQ line into its own dedicated block** — moved AFTER all `box.new()` calls (P&D zones AND S&D zones)
- 2. **Delete-and-redraw pattern** — `line.delete(pdEqLine)` → `line.new()` on every bar ensures the line is ALWAYS the last visual element rendered
- 3. **Width increased:** 4 → 5 (maximum visibility)
- 4. **Color:** `color.white` (pure white, 0% transparency)
- 5. **Style:** `line.style_solid`, `extend.right`
- 6. **EQ Label:** Solid black background (`color.new(color.black, 0)` — 0% transparency), `size.normal`, `label.style_label_left`

Before: EQ line invisible (hidden behind zone fills)
After: EQ line is the TOPMOST visual element on the chart

HUD Navy Blue Background

Property	v17.19 (Before)	v17.20 (After)
Background	<code>color.new(color.black, 40)</code>	<code>color.new(#1a237e, 20)</code> — Deep Navy Blue
Header Row	Same as body	<code>color.new(#0d47a1, 10)</code> — Brighter Navy
Frame Border	<code>color.new(color.white, 80)</code> — faint	<code>color.new(color.white, 60)</code> — visible
Hover Inactive	<code>color.new(color.black, 85)</code>	<code>color.new(#1a237e, 75)</code>

HUD Color Hierarchy Update

Element	v17.19 (Before)	v17.20 (After)
Labels	<code>color.white</code>	<code>color.white</code> (unchanged)
Active Values	<code>color.lime</code>	<code>color.new(color.lime, 0)</code>
Inactive Values	<code>color.new(color.gray, 30)</code>	<code>color.new(color.silver, 0)</code>
Warnings	<code>color.orange</code>	<code>color.new(color.orange, 0)</code>
Invalidated	<code>color.red</code>	<code>color.new(color.red, 0)</code>

Delivery Checklist

#	Task	Status
1	Backup v17.19 → v17.19_backup.pine	✓
2	Create v17.20 working copy	✓
3	Diagnose EQ line z-order issue	✓
4	Move EQ line drawing AFTER all box.new() calls	✓
5	Implement delete-and-redraw pattern for EQ line	✓
6	Increase EQ line width: 4 → 5	✓
7	EQ label: solid black background (0% transparency)	✓
8	HUD background: black → Navy Blue (#1a237e)	✓
9	HUD header row: brighter navy (#0d47a1)	✓
10	HUD frame border: more visible (60% transp)	✓
11	Color hierarchy: silver inactive, explicit 0% transp	✓
12	Update 4 alert JSON payloads to v17.20	✓
13	Update HUD header text to v17.20	✓
14	Update indicator title to v17.20	✓
15	Rename WIP to final .pine file	✓
16	Update app.py to v17.20	✓
17	Update templates/base.html to v17.20	✓
18		✓

#	Task	Status
	Update templates/set-tings.html to v17.20	
19	Update templates/dash-board.html to v17.20	✓
20	Update templates/trades.html to v17.20	✓
21	Update templates/opportunities.html to v17.20	✓
22	Update src/webhook_server/routes.py to v17.20	✓
23	Update src/oie_processor.py to v17.20	✓
24	Update src/decoders.py to v17.20	✓
25	Update src/database.py to v17.20	✓
26	Update src/oie_database.py to v17.20	✓
27	Update schemas/mi-grate_v17_14_oie.sql to v17.20	✓
28	Update tradex-frontend/app/layout.tsx to v17.20	✓
29	Update tradex-frontend/app/page.tsx to v17.20	✓
30	Update tradex-frontend/app/opportunities/page.tsx to v17.20	✓
31	Update tradex-frontend/app/settings/page.tsx to v17.20	✓
32	Plot budget verified: 33/64	✓
33	ZERO logic/filter/alert changes confirmed	✓

#	Task	Status
34	Git commit and push	✓

Modified Files (20 files)

Pine Script (2):

- smc_premium_discount_indicator_v17.20.pine — NEW
- smc_premium_discount_indicator_v17.19_backup.pine — BACKUP

Python Backend (5):

- app.py
- src/webhook_server/routes.py
- src/oie_processor.py
- src/decoders.py
- src/database.py
- src/oie_database.py

HTML Templates (5):

- templates/base.html
- templates/settings.html
- templates/dashboard.html
- templates/trades.html
- templates/opportunities.html

SQL (1):

- schemas/migrate_v17_14_oie.sql

Next.js Frontend (4):

- tradex-frontend/app/layout.tsx
- tradex-frontend/app/page.tsx
- tradex-frontend/app/opportunities/page.tsx
- tradex-frontend/app/settings/page.tsx

TradingView Installation Steps

1. Open TradingView → Pine Script Editor
2. Delete old v17.19 indicator code
3. Paste entire contents of `smc_premium_discount_indicator_v17.20.pine`
4. Click “Add to Chart”
5. **Verify:** EQ line (white, width 5) is NOW VISIBLE on top of P&D zones
6. **Verify:** HUD background is Navy Blue (not black)
7. **Verify:** HUD header shows “SMC P&D v17.20”

Expected Behavior After Installation

- **EQ Line:** Thick white solid line clearly visible across the middle of the P&D zone
- **EQ Label:** “ EQ ” in white text on solid black background, right side of line

- **HUD:** Deep navy blue background with brighter navy header row
- **All existing alerts, logic, and filters:** UNCHANGED